

Air University



Programming Fundamentals (Lab)

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Assignment #03

Code

Row (1-29)

- Used a structure for saving the records of Book store.
- Created a function to add books into the inventory of store.
- Opened a txt file for storing the record in.
- Validated if the txt file is opened or not and displayed an error message.

```
1 #include <iostream>
2 #include <fstream>
3 #include <string>
4 #include <iomanip>
5 using namespace std;
6
7 struct BookStore // Structure to store Books info
8 {
9     int id;
10    string title;
11    string author;
12    float price;
13 };
14
15 void addBook(const BookStore& book) // Function to add books
16 {
17     ofstream outfile("Book Store.txt"); // writing in file
18     if (outfile.is_open()) // checking if file is open
19     {
20         outfile << book.id << "\n" << book.title << "\n" << book.author << "\n" << book.price << "\n";
21         outfile.close();
22         cout << "Book added successfully!\n";
23     }
24     else // showing error
25     {
26         cout << "Error opening file for writing.\n";
27     }
28 }
29
```

Row (30-61)

- Created a function to display books available in the inventory.
- Opened an existing file to reading content from within.
- Validated the file opened and displayed an error message if file is not opened for some reason.
- Set the width to show the content in proper format.
- Used while loop to read from file after the file validation.
- Displayed the output with the pre-set format.
- Closed file outside the while loop.

```
30 void displayBooks() // function to display books in record
31 {
32     ifstream infile("books.txt"); // reading from file
33     if (infile.is_open()) // checking if file is open
34     {
35         // setting width of output
36         cout << left << setw(10) << "Book ID" << setw(30) << "Title"
37         << setw(20) << "Author" << setw(10) << "Price" << "\n";
38         cout << string(70, '-') << "\n";
39
40         while (true) // validating if file is open
41         {
42             BookStore book;
43             infile >> book.id;
44             infile.ignore(); // ignore the newline after the ID
45             if (infile.eof()) break; // Break if end of file is reached
46             getline(infile, book.title);
47             getline(infile, book.author);
48             infile >> book.price;
49             infile.ignore(); // Ignore the newline after the price
50
51             // displaying the output
52             cout << left << setw(10) << book.id << setw(30) << book.title
53             << setw(20) << book.author << setw(10) << book.price << "\n";
54         }
55         infile.close(); // closing file
56     }
57     else // showing error if occurs
58     {
59         cout << "Error opening file for reading.\n";
60     }
61 }
```

Row (63-96)

- Created a function to search book using ID of book within the inventory of store.
- Opened file to read from.
- Used while loop after validating the file opening.
- After reading the records from the file, compared the

```
63 void searchBook(int id) // function to search book from inventory
64 {
65     ifstream infile("Book Store.txt"); // reading from file
66     if (infile.is_open()) // checking if file is open
67     {
68         while (true) // validating if file is open
69         {
70             BookStore book;
71             infile >> book.id;
72             infile.ignore(); // Ignore the newline after the ID
73             if (infile.eof()) break; // Break if end of file is reached
74             getline(infile, book.title);
75             getline(infile, book.author);
76             infile >> book.price;
77             infile.ignore(); // Ignore the newline after the price
78
79             if (book.id == id) // comparing the ID
80             {
81                 // displaying the book found
82                 cout << "Book found:\n";
83                 cout << "Book ID: " << book.id << "\n" << "Title: " << book.title << "\n"
84                 << "Author: " << book.author << "\n" << "Price: " << book.price << "\n";
85                 infile.close(); // closing the file
86                 return; // returning to main
87             }
88         }
89         // showing that book with searched ID does not exists
90         cout << "Book with ID " << id << " not found.\n";
91         infile.close(); // closing file
92     }
93     else // showing error if any
94     {
95         cout << "Error opening file for reading.\n";
96     }
```

ID received from user with the ID's of books already in the inventory to find and display the book.

- Showed a message indicating that the book does not exist in the inventory.
- Displayed a file opening error if file is not opened for some reason.

Row (99-131)

- Main Function.
- Declared a variable to store input from user.
- Displayed the task list.
- Used switch to handle the task which user wants to perform.
- Asked information about new book which the user wants to add in inventory.
- Called the function to save the new book in inventory.

```
99 int main() // main function
100 {
101     int choice; // variable for switch
102     do
103     { // inventory Menu
104         cout << "\nBook Store Inventory System\n";
105         cout << "1. Add Book\n";
106         cout << "2. Display All Books\n";
107         cout << "3. Search Book by ID\n";
108         cout << "4. Exit\n";
109         cout << "Enter your choice: ";
110         cin >> choice;
111
112         switch (choice)
113         {
114             case 1:
115             {
116                 BookStore newBook;
117
118                 cout << "Enter Book ID: ";
119                 cin >> newBook.id;
120                 cin.ignore(); // Ignore the newline after the ID
121
122                 cout << "Enter Title: ";
123                 getline(cin, newBook.title);
124
125                 cout << "Enter Author: ";
126                 getline(cin, newBook.author);
127
128                 cout << "Enter Price: ";
129                 cin >> newBook.price;
130                 addBook(newBook); // calling function to add new books
131                 break;
132             }
```

Row (133-157)

- Called the function of displaying records in case 2.
- Asked to user to enter ID of book they want to search and then called the function to search the book within the inventory.
- Exited the program if user enter 4.
- Prompted the user to enter valid input.

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case 2:
    displayBooks(); // calling function to display books in inventory
    break;
case 3:
{
    int searchId;

    cout << "Enter Book ID to search: ";
    cin >> searchId;

    searchBook(searchId); // calling function to search books from inventory
    break;
}
case 4:
    cout << "Exiting the program.\n";
    break;
default:
    cout << "Invalid choice. Please try again.\n";
    break;
}
while (choice != 4);

return 0;
```

Output

```
D:\University\Programming Fundamentals (Lab)\Assignments\Assignment 3\Assignment 3.exe

Book Store Inventory System
1. Add Book
2. Display All Books
3. Search Book by ID
4. Exit
Enter your choice: 1
Enter Book ID: 2001
Enter Title: CyberSecurity
Enter Author: SecretCoder
Enter Price: 5000
Book added successfully!

Book Store Inventory System
1. Add Book
2. Display All Books
3. Search Book by ID
4. Exit
Enter your choice: 2
Book ID      Title                      Author          Price
-----
2001         CyberSecurity              SecretCoder     5000

Book Store Inventory System
1. Add Book
2. Display All Books
3. Search Book by ID
4. Exit
Enter your choice: 3
Enter Book ID to search: 2000
Book with ID 2000 not found.
```

```
Book Store Inventory System
1. Add Book
2. Display All Books
3. Search Book by ID
4. Exit
Enter your choice: 3
Enter Book ID to search: 2001
Book found:
Book ID: 2001
Title: CyberSecurity
Author: SecretCoder
Price: 5000

Book Store Inventory System
1. Add Book
2. Display All Books
3. Search Book by ID
4. Exit
Enter your choice: 5
Invalid choice. Please try again.

Book Store Inventory System
1. Add Book
2. Display All Books
3. Search Book by ID
4. Exit
Enter your choice: 4
Exiting the program.

-----
Process exited after 40.18 seconds with return value 0
Press any key to continue . . .
```

Thank You for Your Time!

THE END